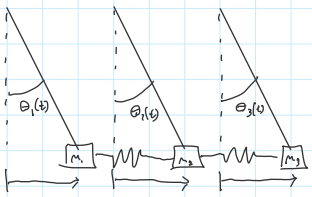
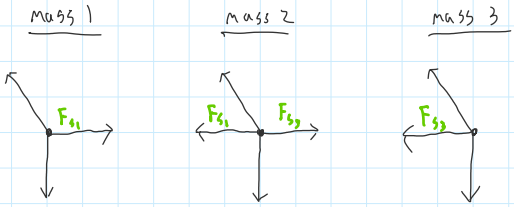
Triple mass coupled pendulum

Thursday, March 5, 2026 8:20 AM

With only a few small modifications to our governing equations we can model a 3 mass system, and maybe even an n mass system....





$$F_p = -\frac{mg}{r}\sin\theta \approx -\frac{mg}{r}u(t)$$

$$F_{net\,i} = F_p - F_{s_i} + F_{s_{i-1}} \; ; \; F_{s_i} = -k(u_{i+1}(t) - u_i(t))$$

$$\Rightarrow \quad \ddot{u}_i(t) = -\frac{g}{r}u_i(t) + \frac{k}{m}\left(u_{i+1}(t) - u_i(t)\right) - \frac{k}{m}\left(u_i(t) - u_{i-1}(t)\right)$$

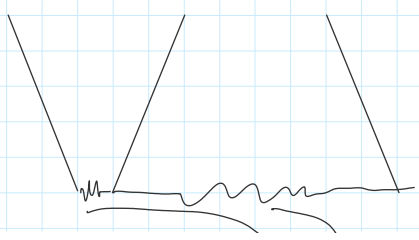
for $i \in \{1, 2, 3\}$

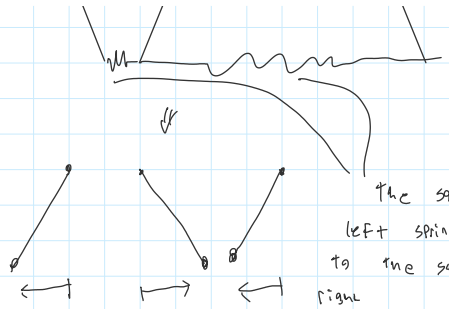
$$\ddot{u}_1(t) = -\frac{g}{r}u_1(t) - \frac{k}{m}u_1(t) + \frac{k}{m}u_2(t)$$

$$\ddot{u}_2(t) = -\frac{g}{r}u_2(t) - \frac{k}{m}u_2(t) + \frac{k}{m}u_3(t) - \frac{k}{m}u_2(t) + \frac{k}{m}u_1(t)$$

$$\ddot{u}_3(t) = -\frac{g}{r}u_3(t) - \frac{k}{m}u_3(t) + \frac{k}{m}u_2(t)$$

$$A = \begin{bmatrix} -\frac{g}{r} - \frac{k}{m} & \frac{k}{m} & 0 \\ \frac{k}{m} & -\frac{g}{r} - \frac{2k}{m} & \frac{k}{m} \\ 0 & \frac{k}{m} & -\frac{g}{r} - \frac{m}{m} \end{bmatrix}_{3\times 3}$$





The squish of the left spring is equal to the stretch of the right

$\Rightarrow$

$\cancel{U_1 + U_3 + 2U_2 = 0}$

$$\boxed{\begin{aligned} &X_1: U_1 + U_3 = -\tfrac{1}{2}U_2 \\ &X_2: U_1 = U_2 = U_3 \\ &X_3\ U_1 = -U_3 \quad U_2 = 0 \end{aligned}}$$

eigenvectors of 3 mass system

Eigenvalue analysis:

Let: $U = \begin{bmatrix} U_1(t) \\ U_2(t) \\ U_3(t) \end{bmatrix}$ $\ddot{U} = \begin{bmatrix} \ddot{U}_1(t) \\ \ddot{U}_2(t) \\ \ddot{U}_3(t) \end{bmatrix}$ $A = \begin{bmatrix} -\frac{g}{r} - \frac{k}{m} & \frac{k}{m} & 0 \\ \frac{k}{m} & -\frac{g}{r} - \frac{2k}{m} & \frac{k}{m} \\ 0 & \frac{k}{m} & -\frac{g}{r} - \frac{k}{m} \end{bmatrix}$

& $\ddot{U} = A \cdot U$

assume $U = Xe^{st}$: $X \in \mathbb{R}^3$

$\Rightarrow \ddot{U} = Xs^2e^{st}$

$\Rightarrow \quad Xs^2e^{st} = AXe^{st}$

$\Rightarrow \quad e^{st}(AX - \lambda X) = 0 \quad : \lambda = s^2$

$\Rightarrow \quad AX - \lambda X = 0 \quad : \lambda = s^2$

$X \neq 0 \quad \Rightarrow \quad X(A - I_3\lambda) = 0 \quad : \lambda = s^2$

$\Rightarrow \quad A - I_3\lambda = 0_3 \quad : \lambda = s^2$

$$\Rightarrow \begin{bmatrix} -\frac{g}{r} - \frac{k}{m} - \lambda & \frac{k}{m} & 0 \\ \frac{k}{m} & -\frac{g}{r} - \frac{2k}{m} - \lambda & \frac{k}{m} \\ 0 & \frac{k}{m} & -\frac{g}{r} - \frac{k}{m} - \lambda \end{bmatrix}$$

$$\left(-\frac{g}{r} - \frac{k}{m} - \lambda\right)\left[\left(-\frac{g}{r} - \frac{k}{m} - \lambda\right)\left(-\frac{g}{r} - 2\frac{k}{m} - \lambda\right) - 2\frac{k^2}{m^2}\right] = 0$$

$-\frac{g}{r} - \frac{k}{m} - \lambda = 0$ & $\left(-\frac{g}{r} - \frac{k}{m} - \lambda\right)\left(-\frac{g}{r} - 2\frac{k}{m} - \lambda\right) - 2\frac{k^2}{m^2} = 0$

$$\Rightarrow \begin{bmatrix} \frac{k}{m} & -\frac{g}{r}-\frac{2k}{m}-\lambda & \frac{k}{m} \\ 0 & \frac{k}{m} & -\frac{g}{r}-\frac{k}{m}-\lambda \end{bmatrix}$$

$$-\frac{g}{r}-\frac{k}{m}-\lambda=0 \quad \& \quad \left(-\frac{g}{r}-\frac{k}{m}-\lambda\right)\left(-\frac{g}{r}-2\frac{k}{m}-\lambda\right)-2\frac{k^2}{m^2}=0$$

$$\lambda^2+\lambda\left(2\frac{g}{r}+3\frac{k}{m}\right)+\frac{g^2}{r^2}+2\frac{k^2}{m^2}+3\frac{gk}{rm}=0$$

$$-\left(\frac{g}{r}+\frac{k}{m}+\lambda\right)^2\left(\frac{g}{r}+\frac{2k}{m}+\lambda\right)-2\left(-\frac{g}{r}-\frac{k}{m}-\lambda\right)\left(\frac{k}{m}\right)^2=0$$

$$\lambda^2+\lambda\left(2\frac{g}{r}+3\frac{k}{m}\right)+\frac{g^2}{r^2}+3\frac{gk}{rm}=0$$

$$\cancel{-\left(\frac{g^2}{r^2}+\frac{k^2}{m^2}+\lambda^2+2\frac{gk}{mr}+2\lambda\frac{g}{r}+2\lambda\frac{k}{m}\right)\left(\frac{g}{r}+\frac{2k}{m}+\lambda\right)+2\frac{k^2g}{m^2r}+2\frac{k^3}{m^3}+2\lambda\frac{k^2}{m^2}=0}$$

$$-\frac{g^3}{r^3}-2\frac{kg^2}{mr^2}-\lambda\frac{g^2}{r^2}-\frac{gk^2}{rm^2}-2\frac{k^3}{m^3}-\lambda\frac{k^2}{m^2}-\lambda^2\frac{g}{r}-2\lambda^2\frac{k}{m}-\lambda^3-2\frac{g^2k}{r^2m}-\cancel{4}\,\frac{gk^2}{rm^2}-2\lambda\frac{gk}{rm}\ldots$$

$$-2\lambda\frac{g^2}{r^2}-4\lambda\frac{gk}{rm}-2\lambda^2\frac{g}{r}-2\lambda\frac{gk}{rm}-\cancel{4}\,\lambda\frac{k^2}{m^2}-2\lambda^2\frac{k}{m}+2\frac{k^2g}{m^2r}+2\frac{k^3}{m^3}+2\lambda\frac{k^2}{m^2}=0$$

$$\Rightarrow -\lambda^3-\lambda^2\left(+4\frac{k}{m}+3\frac{g}{r}\right)-\lambda\left(2\frac{k^2}{m^2}+3\frac{g^2}{r^2}+8\frac{gk}{rm}\right)-\frac{g^3}{r^3}-4\frac{g^2k}{r^2m}-3\frac{gk^2}{rm^2}=0$$

Let $A=\frac{g}{r}$ & $B=\frac{k}{m}$

$$\Rightarrow \boxed{\lambda^3+\lambda^2(4B+3A)+\lambda(2B^2+3A^2+8AB)+A^3+4A^2B+3AB^2=0}$$

Would ya look at that, it's a cubic.

this is just Generally a horrible way to do Algebra.

If your eq is allready factored, you should only unfactorize it as a last resort. if it get's that far just use Newtons method.

Since we allready know $\lambda=-\frac{g}{r}$ should be an eigen value

$$\Rightarrow \cancel{\left(-\frac{g}{r}\right)^2}+\left(-\frac{g}{r}\right)\cancel{\left(2\frac{g}{r}+3\frac{k}{m}\right)}+\cancel{\frac{g^2}{r^2}}+\cancel{3\frac{gk}{rm}}=0 \quad ✓$$

we can factor out $\left(\lambda+\frac{g}{r}\right)$

$$\left(\lambda+\frac{g}{r}\right)\overline{\left(\lambda+\left(\frac{g}{r}+3\frac{k}{m}\right)\right)}$$

$$\lambda^2+\lambda\left(2\frac{g}{r}+3\frac{k}{m}\right)+\frac{g^2}{r^2}+3\frac{kg}{rm}$$

$$-\ \lambda^2+\lambda\left(\frac{g}{r}\right)$$

$$\lambda\left(\frac{g}{r}+3\frac{k}{m}\right)+\frac{g^2}{r^2}+3\frac{kg}{rm}$$

$$-\ \lambda\left(\frac{g}{r}+3\frac{k}{m}\right)+\frac{g^2}{r^2}+3\frac{kg}{rm}$$

$$0$$

$$\Rightarrow \left(\lambda+\frac{g}{r}+3\frac{k}{m}\right)=0$$

$$\Rightarrow \lambda=-\frac{g}{r}-3\frac{k}{m}$$

$$\boxed{\lambda=-\frac{g}{r},\ -\frac{g}{r}-3\frac{k}{m},\ -\frac{g}{r}-\frac{k}{m}}$$